



**CALICUT UNIVERSITY – FOUR-YEAR UNDER
GRADUATE PROGRAMME (CU-FYUGP)**

BSc PHYSICS HONOURS

Programme	B.Sc. Physics Honours				
Course Title	NANOSCIENCE AND TECHNOLOGY				
Type of Course	Major Elective (SPECIALIZATION I: MATERIALS SCIENCE)				
Semester	VI				
Academic Level	300 - 399				
Course Details	Credit	Lecture per week	Tutorial per week	Practical per week	Total Hours
	4	4	-	-	60
Pre-requisites	PHY5EJ302(1)- Materials Science				
Course Summary	This Nanoscience and Technology aims to provide students with a solid foundation in the principles, techniques, and applications of nanotechnology, preparing them for careers in research and industry.				

Course Outcomes (CO):

CO	CO Statement	Cognitive Level*	Knowledge Category#	Evaluation Tools used
CO1	Understanding of Nanoscale Phenomena, the unique properties and behaviors of materials at the nanoscale.	U	F	Instructor-created exams / Quiz
CO2	Understand the science of nanomaterials: including quantum effects, surface phenomena, and size-dependent properties.	U, Up	F	Instructor-created exams / Quiz

CO3	Understand the knowledge about the type of nanomaterials and how the size effect affects the transport properties in nanomaterials	U, An	F	Instructor-created exams / Quiz
CO4	Knowledge of Nanofabrication Techniques: Students should learn about various techniques used to fabricate nanostructures and nanomaterials, such as top-down and bottom-up approaches.	U	F	Instructor-created exams / Quiz
CO5	To familiar with a range of characterization techniques used to analyze nanomaterials and nanostructures using conventional and advanced techniques.	U, An	F	Instructor-created exams / Quiz
CO6	Research Skills: Depending on the level of the course, students may develop research skills through laboratory work, independent projects, or literature reviews.	U, Ap	F	Instructor-created exams / Quiz
* - Remember (R), Understand (U), Apply (Ap), Analyse (An), Evaluate (E), Create (C) # - Factual Knowledge(F) Conceptual Knowledge (C) Procedural Knowledge (P) Metacognitive Knowledge (M)				

Detailed Syllabus:

Module	Unit	Content	Hrs (48 +12)	Marks (70)
I	INTRODUCTION OF NANOSCALE SCIENCE		15	22
	1	Introduction to the nanoscale, Size effects in small systems, Quantum behaviors of the nanometric world ¹	2	
	2	Applications of Schrodinger equation - infinite potential well, potential step, potential box; trapped particle in 3D (nanodot), electron trapped in 2D plane (nanosheet), electrons moving in 1D (nanowire, nanorod, nanobelt) ¹	5	
	3	Quantum confinement effect in nanomaterials. Electron confinement ¹	2	

	4	Density of states: Density of States for a Zero Dimensional (0D) Solid	3	
	5	Special nanomaterials: Density of States in a Two-Dimensional (2D) Potential Box	2	
	6	Thin Film, Density of States for a Particle in a Three-Dimensional Box	1	
	Chapter V of Book 1 and Sections 1.5, 1.5.2 of Book 2			
II	NANOSTRUCTURES		10	15
	7	Nanostructures: Zero, One Two, and Three-dimensional nanostructures,	3	
	8	some special nanostructures: Carbon nanomaterials, Fullerenes, Carbon Nanotubes (CNTs), Types of carbon nanotubes, Graphene	4	
	9	Metal-Organic Frameworks (MOF), Core-Shell Particles, Metamaterials, Bioinspired Materials.	3	
	Sections 11.2, 11.2.1, 11.2.2, 11.2.3, 11.2.6, 11.7, 11.8, 11.9 of Book 2			
III	SYNTHESIS OF NANOMATERIALS		10	15
	10	Mechanical Methods: High Energy Ball Milling	1	
	11	Physical Vapour Deposition with Consolidation, Laser Vaporization (Ablation)	1	
	12	Chemical Vapour Deposition (CVD), Ion Beam Techniques (Ion Implantation, Molecular Beam Epitaxy (MBE).	2	
	13	Synthesis of Nanomaterials—II (Chemical Methods): Sol-Gel Method, Hydrothermal Synthesis,	2	
	14	Sonochemical Synthesis, Microwave Synthesis	2	
	15	Self-Assembly: Basic Mechanism and Self Assembly of Nanoparticles Using Organic Molecules.	2	
	Sections 3.1, 3.2.1, 3.3, 3.3.1, 3.3.3, 3.4, 3.4.1, 3.4.2, 3.5, 3.7, 3.8, 4.8, 4.9, 4.10, 4.11, 6.1, 6.3.1 of Book 2			
IV	ANALYSIS TECHNIQUES		13	18
	16	Analysis Techniques: Optical Microscope: Confocal Microscope	2	
	17	Electron Microscopes: Scanning Electron Microscope	2	
	18	Transmission Electron Microscope (TEM)	2	

	19	Scanning Probe Microscopes (SPM), Atomic Force Microscope, Scanning Probe Microscopes (SPM)	2	
	20	XRD and diffraction from different types of samples	1	
	21	Diffraction from Nanoparticles.	2	
	22	X-Ray Ultra Violet Photoelectron Spectroscopies	2	
	Sections 7.2 7.2.2, 7.3, 7.4, 7.4.1, 7.4.2, 7.5.1, 7.5.4, 7.5.6, 7.6.8 of Book 2			
V	OPEN ENDED MODULE		12	
		Applications of Nanomaterials: Organic Photovoltaic cells, Fuel Cell, Hydrogen Generation and Storage, Photo Electrochemical Cells (PEC), Hybrid Energy Cells, Automobiles, Textiles, Sports and Toys, Cosmetics, Medical field, Space, Defense and Engineering.		
		Synthesis of nanoparticles and analysis using XRD, SEM, TEM, Optical Methods, etc		
	Sections 12.2, 12.2.2, 12.2.3-12.2.6, 12.3-12.7, 12.10, 14.4 of Book 2			
Books and References:				
1. Introduction to Nanoscience and Nanotechnology, Chattopadhyaya and A. N. Banerjee, Publisher: PHI Learning and Private Limited (Book 1)				
2. Nanotechnology: Principles and Practices, Sulabha K. Kulkarni, Springer, 3 rd Edition (Book 2)				

Mapping of COs with PSOs and POs :

	PSO 1	PSO 2	PSO 3	PSO4	PS O5	PS O6	PO1	PO2	PO3	PO4	PO5	PO 6	PO 7
CO 1	3	2	3	2	3	3	3	2	3	2	3	3	3
CO 2	3	2	3	2	3	3	3	2	3	2	3	3	3
CO 3	3	2	3	2	3	3	3	2	3	2	3	3	3
CO 4	3	2	3	2	3	3	3	2	3	2	3	3	3
CO 5	3	2	3	2	3	3	3	2	3	2	3	3	3
CO 6	3	2	3	2	3	3	3	2	3	2	3	3	3

Correlation Levels:

Level	Correlation
0	Nil
1	Slightly / Low
2	Moderate / Medium
3	Substantial / High

Assessment Rubrics:

- Quiz / Discussion / Seminar
- Internal Theory/Practical Exam
- Assignments /Viva
- End Semester Exam (70%)

Mapping of COs to Assessment Rubrics

	Internal Theory /Practical Exam	Assignme nt /Viva	Practical Skill Evaluation	End Semester Examinations
CO 1	✓	✓		✓
CO 2	✓	✓		✓
CO 3	✓	✓		✓
CO 4	✓	✓		✓
CO 5	✓	✓		✓
CO 6		✓	✓	